Design and Development of IoT-Based Smart Maldive Fish Processing System

COMMON INTEGRATED SOLUTION REPORT

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DECLARATION AND SUPERVISOR CERTIFICATION

Candidate declaration. We declare that this common report is our own work and does not incorporate, without acknowledgement, material previously submitted for a degree or diploma at any university or institute of higher learning. To the best of our knowledge, it contains no material previously published or written by another person except where acknowledgement is made in the text. We grant the Sri Lanka Institute of Information Technology the non-exclusive right to reproduce and distribute this report, in whole or in part, in print, electronic, or other media, while retaining the right to use the content in future work.

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Supervisor certification. The above candidates carried out this research project under my supervision.

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ABSTRACT

Maldive fish processing involves interconnected stages, such as raw fish quality assessment, boiling and salting drying and final product inspection. The current process depends largely on quality checks and limited monitoring of processing and environmental conditions. This can cause quality decisions, inefficient processing and delayed detection of conditions that may affect product quality.

The research presents an AI- and IoT-based smart Maldive fish processing system that combines four complementary components: AI-driven raw fish quality assessment and standardization smart optimization of boiling and salt distribution IoT-based environmental monitoring during drying and AI-based multi-sensor quality evaluation of dried Maldive fish.

The proposed system integrates computer vision, AI/ML models, ESP32-based sensing, gas sensing, process monitoring and a centralized dashboard to support continuous data-driven decision making throughout the processing lifecycle.

Experimental evaluation in the project research showed 95.8 % mAP@0.5 for raw fish freshness assessment a 27.9 % ± 1.4 % reduction in boiling cycle time a decrease in deep-tissue salt variation from 18.6 % ± 2.1 % to 5.2 % ± 0.8 %, 99.6 % MQTT telemetry delivery during drying and 94.0 % accuracy for dried-fish quality classification.

The integrated solution shows that it is feasible to combine AI, IoT sensing, process optimization and digital monitoring within a Maldive fish processing workflow. The system can support consistent quality assessment, improved process efficiency, real-time environmental monitoring and faster identification of quality-related risks. This provides a foundation for commercial-scale deployment and further validation.

Keywords: [three to five keywords, separated by commas]

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LIST OF ABBREVIATIONS

[Guidance - delete before submission: List abbreviations alphabetically. Define each abbreviation at first use in the main text even when it appears here.]

Abbreviation	Meaning
[ABC]	[Full term]
[ABC]	[Full term]
[ABC]	[Full term]

1 INTRODUCTION AND PROJECT CONTEXT

1.1 Background and context

Maldive fish processing is a key traditional fish-processing activity in Sri Lanka. The quality of the finished product relies on linked steps, such as raw fish selection, boiling, salting, drying and final quality assessment. This research aims to improve that workflow by adding Artificial Intelligence (AI) Internet of Things (IoT) computer vision, sensor technologies and data-driven process optimization.

The project research shows that normal Maldive fish processing depends heavily on inspection and old practices. People usually judge raw fish quality by looking at it and the cooking and drying conditions are not always watched continuously. These habits can lead to quality choices, waste of resources and late detection of environmental or product problems.

The proposed project tackles these problems with a processing system made of four parts. The first part uses AI-powered computer vision to check and set the quality of raw fish. The second part aims to improve boiling and salt distribution by using sensors and smart control. The third part gives IoT-based monitoring while the fish dries watching the factors that affect drying. The fourth part mixes AI image analysis with sensors to judge the quality of the finished dried Maldive fish.

By linking these four parts data can move through all stages of production instead of treating each step alone. Sensor readings and AI predictions can give decisions all along the process. A central monitoring system also helps us see process conditions and quality issues clearly.

Overall the project shows how AI and IoT can modernize fish processing while still meeting the practical needs of the current workflow. The system gives a tech base to raise quality consistency, monitor processes boost efficiency and make data-based decisions, in fish production.

1.2 Stakeholders and user needs

The smart Maldive fish processing system touches stakeholders who are directly or indirectly involved in processing, quality checks and monitoring. The main groups we look at are fish producers, fish processing factories, fish quality inspectors, small and medium fish-processing businesses and seafood exporters. Consumers are also stakeholders because better quality checks and monitoring can give them more consistent products.

Table 1 - Stakeholders and Their Main Needs

Stakeholder	Main Needs
Maldive fish producers	Efficient processing, consistent product quality, reduced manual errors, and improved monitoring of processing conditions
Fish processing factories	Automated quality assessment, process optimization, environmental monitoring, and centralized information
Fish quality inspectors	Faster and more consistent quality evaluation with AI-assisted inspection
Small and medium-scale businesses	Affordable monitoring and quality-assessment support that can reduce dependence on continuous manual inspection
Seafood exporters	Consistent product quality, reliable quality records, and improved confidence in processing standards
Consumers	Consistent-quality Maldive fish products and reduced risk of poor-quality products reaching the market

The needs of Fish quality inspectors, Small and medium-scale businesses, Seafood exporters and Consumers are addressed through the integration of the four project components. The raw fish quality assessment component supports producers and Fish quality inspectors by providing AI-assisted classification of fish. The boiling and salt-distribution optimization component supports processing operations by

monitoring process parameters and assisting with more consistent processing conditions.

During drying the IoT-based environmental monitoring component supplies information about environmental conditions that may affect the drying process. This allows processing personnel to identify unsuitable conditions and respond to environmental changes. Finally, the multi-sensor dried-fish quality evaluation component supports product inspection by combining image-based quality assessment with sensor-based information related to product condition.

The centralized monitoring and dashboard concept offers an interface for viewing relevant information from different stages of the production process. Therefore the proposed system is designed to automate inspection tasks and to provide Fish quality inspectors, Small and medium-scale businesses, Seafood exporters and Consumers with a more connected and data-driven approach to Maldive fish processing.

Overall, fish quality inspectors, medium-scale businesses, Seafood exporters and Consumers require quality consistency, timely information, reduced dependence on manual inspection, improved process efficiency and better visibility of processing conditions. These requirements form the basis for the integrated architecture and functional objectives described in the following chapters.

1.3 Alignment with the Sustainable Development Goals

The proposed AI- and IoT-based smart Maldive fish processing system can support United Nations Sustainable Development Goals (SDGs). It focuses on improving processing efficiency supporting innovation and promoting more responsible production practices. The project mainly aligns with SDG 9: Industry, Innovation and Infrastructure and SDG 12: Responsible Consumption and Production.

SDG 9 – Industry, Innovation and Infrastructure The project contributes to SDG 9 objectives by applying emerging technologies to a traditional fish-processing workflow. The proposed system integrates Artificial Intelligence, Internet of Things technologies, computer vision, sensor-based monitoring and data-driven process optimization across stages of Maldive fish production.

The four integrated components provide support for raw fish quality assessment, boiling and salt-distribution optimization, drying-environment monitoring and final dried-fish quality evaluation. This integration shows how digital technologies can improve monitoring and management of an existing production process. The system can therefore provide a foundation for development of smart processing infrastructure within the Maldive fish-processing sector.

SDG 12 – Responsible Consumption and Production The project aligns with SDG 12 by contributing to controlled and efficient production practices. Monitoring processing and environmental conditions helps identify conditions at different production stages while AI-assisted quality assessment supports more consistent quality decisions.

The boiling and salt-distribution optimization component is intended to support efficient use of processing resources. Environmental monitoring during drying gives information that assists operators in responding to conditions that may negatively affect the process. In addition final product quality assessment can identify lower-quality products before distribution.

Therefore the proposed system supports efficient resource use, improved process control reduced quality-related losses and more consistent production practices. These contributions show the projects intended alignment with SDG 12 than a claim of direct measurable SDG impact.

1.4 Purpose and scope of the common report

The purpose of this common report is to present the overall problem, integrated solution, system architecture, component interactions, implementation approach and overall evaluation of the proposed Design and Development of IoT-Based Smart Maldive Fish Processing System. Unlike component-level research reports this report focuses on how the four research components work as one integrated solution, for improving Maldive fish processing and quality management.

The common report brings together the four components of the project: Raw Fish Quality Assessment and Standardization Smart Optimization of Boiling and Salt Distribution Intelligent IoT-Based Multi-Parameter Environmental Monitoring and

AI-Based Multi-Sensor Quality Evaluation of Dried Maldive Fish. It describes how these components contribute to stages of the overall processing workflow and how information, sensor data, AI predictions and monitoring outputs connect to support end-to-end decision making.

The scope of this report covers topics. First it defines the processing problem. Next it lists the needs of stakeholders. Then it shows the integrated system architecture. It also explains how the component interfaces work together and which dependencies exist. The report discusses the shared technology infrastructure and the overall implementation of the system. Integration testing, overall evaluation, commercialization considerations, sustainability and future development directions are also covered. Special focus is placed on how information flows among the four components and how they work together to assess quality optimize the process monitor the environment and evaluate the product.

Each of the four theses presents research and development evidence for its component. Those reports include a literature review, a research gap, the methodology, data collection, model development, implementation, testing and detailed results for that component. Because of this the current report does not repeat all details for each component. Instead it pulls together the findings and shows how the separate contributions build a cohesive AI- and IoT-based Maldive fish processing system.

The overall scope is limited to the research prototype and its tested processing workflow. Commercial-scale deployment, long-term industrial validation and large-scale operational adoption are seen as future directions. They are not claimed as finished results of the project.

2 OVERALL PROBLEM DEFINITION AND FOUR SUBPROBLEMS

2.1 Overall problem definition

The main problem that this research tackles is the absence of a single data-driven system that keeps quality consistent and controls important processing conditions during every stage of Maldive fish production. The current workflow contains linked stages: raw fish assessment, boiling and salting drying and final product evaluation. These stages usually suffer from inspection, limited monitoring, variable conditions and separate quality checks that are spread across stages.

At the stage manual assessment can lead to subjective and inconsistent decisions about fish quality. In boiling and salting wrong processing times and uneven salt distribution can damage product quality. Reduce efficiency. In drying environmental factors such as temperature, humidity, airflow and gas levels affect how well the fish dries and how stable the product remains. In the stage relying only on visual inspection may not give enough details about product condition or spoilage.

The overall issue is not one processing step. Each stage depends on the one so problems that start early can affect later stages and the final product. The paper shows that we need a framework that blends raw-material inspection, process optimization, environmental monitoring and final product assessment. It should not treat these tasks as problems.

2.2 Root causes and overall research gap

Several connected factors cause the problem. First, we still rely on quality inspection. Human judgment depends on each person’s experience, which can cause differences in quality classification and slow down inspection.

The second cause is the monitoring and tuning of boiling and salting parameters. The study shows that boiling long can break muscle fibers and reduce yield while boiling too little can lead to uneven salt spread. Thus, we must consider processing time, batch traits, temperature and salt level together not one by one.

The third cause is the absence of multi-parameter monitoring during drying. Drying depends on the environment, temperature, humidity, airflow and gas conditions. Therefore, the research system tracks environmental factors instead of just one.

The fourth cause is the use of extra sensors for final product checking. Pictures give clues about dried fish quality, but sensors add data about gas levels and product stability. So, the research mixes image assessment with MQ-135 sensor data and shelf-life estimates.

2.3 Subproblem and component mapping

The overall problem splits into four yet connected subproblems. Each component tackles a stage of Maldive fish processing and supplies data that helps the next stages.

Table 2 - Subproblem and Component Mapping

Subproblem	Corresponding Component	Main Focus	Relationship with Other Components
SP1: Inconsistent assessment of raw fish quality	**Component 1 – AI-Driven Raw Fish Quality Assessment and Standardization**	Assessing and classifying raw fish before processing	Provides raw-material quality information that can support subsequent processing decisions
SP2: Inefficient boiling and inconsistent salt distribution	**Component 2 – Smart Optimization of Boiling and Salt Distribution**	Monitoring and optimizing boiling and salting conditions	Uses raw-material/process information and produces a more controlled input for drying
SP3: Lack of continuous monitoring of drying conditions	**Component 3 – Intelligent IoT-Based Multi-Parameter Environmental Monitoring**	Monitoring temperature, humidity, airflow, and gas-related conditions	Monitors the environment affecting the drying stage and provides alerts when conditions become unsuitable
SP4: Incomplete final dried-fish	**Component 4 – AI-Based Multi-Sensor Quality Evaluation**	Combining visual classification and sensor-based	Provides the final quality evaluation after processing and drying

quality assessment	**of Dried Maldive Fish**	quality/shelf-life assessment	

2.4 Overall aim and group objectives

Overall Aim

The overall aim of the research is:

To design and develop an AI- and IoT-based smart Maldive fish processing system that improves quality consistency, processing efficiency, environmental monitoring and data-driven decision-making throughout the Maldive fish processing lifecycle.

This aim matches the project goal of mixing AI, computer vision, IoT sensing and real-time monitoring into one processing framework.

Group Objectives

The overall aim is covered by the following group objectives.

1 – Raw Fish Quality Assessment

To develop an AI-assisted visual inspection that can assess and classify raw fish quality before processing.

Mapped component: Component 1

Evaluation evidence: AI classification performance using model evaluation measures **and test data.**

2 – Boiling and Salt-Distribution Optimization

To create an intelligent process-monitoring and optimization tool that controls boiling conditions and improves salt-distribution consistency.

Mapped component: Component 2

Evaluation evidence: boiling/process efficiency, salt-distribution variation, processing conditions and related process measurements.

Objective 3 – Drying Environment Monitoring

To build an IoT-based monitoring system that continuously measures critical drying-environment conditions and raises alerts for unsuitable conditions.

Mapped component: Component 3

Evaluation evidence: sensor measurements, telemetry reliability response time, environmental-condition detection and drying performance.

Objective 4 – Dried-Fish Quality Evaluation

To develop a multi-sensor quality evaluation that combines AI image classification with sensor measurements for final dried-fish quality assessment and shelf-life decision support.

Mapped component: Component 4

Evaluation evidence: image-classification performance, sensor readings, quality classification and shelf-life estimation.

Objective 5 – Integrated Monitoring and Decision Support

To unite the four components through a common monitoring architecture that supports data exchange, visualization, alerts and data-driven decision-making across the processing lifecycle.

Mapped components: Components 1–4

Evaluation evidence: component integration testing, data communication, end-to-end workflow testing, dashboard operation and system-level evaluation.

Table 3 - Objective to Component to Evidence Mapping

Group Objective	Component(s)	Main Evidence
Raw fish quality assessment	Component 1	Classification performance and test results
Boiling and salt optimization	Component 2	Processing efficiency and salt-distribution measurements
Drying-environment monitoring	Component 3	Sensor accuracy, telemetry reliability, alerts, and drying performance

Final dried-fish quality evaluation	Component 4	Classification performance, sensor measurements, and shelf-life-related outputs
Integrated decision support	Components 1–4	Integration, communication, dashboard, and end-to-end testing

2.5 Scope assumptions and constraints

The scope of the proposed system covers the major quality and monitoring activities from raw fish intake to final dried Maldive fish evaluation. The system is meant to give help and monitoring not to replace human operators entirely in the processing environment.

Scope

The integrated solution includes:

1. AI-assisted raw fish quality assessment.
2.. Optimization of boiling and salt-distribution parameters.
3. IoT-based monitoring of conditions.
4. AI-based visual assessment of dried fish.
5. Sensor-based monitoring related to quality and product stability.
6.. Near-real-time data transmission and monitoring.
7. Visualization and decision-support functionality.
8. Integration of the four components within the processing workflow.

The research paper describes evaluation using production batches and the use of AI models, IoT sensing and process measurements to test the proposed approach.

Assumptions

The integrated solution is developed under the assumptions:

- Required sensors and processing equipment are available and functioning correctly.

- Sensor readings are reliable, after calibration and preprocessing.
- The AI models are. Evaluated using representative datasets for the classification tasks they are intended to perform.
- Internet or local network connectivity must be available for transmitting IoT data when required.
- Processing operators must be available to respond to system notifications and recommendations.
- The measured environmental and process parameters provide indicators of the processing conditions that matter.

Constraints

The project is subject to practical constraints:

- The performance of AI models depends on the quality, size and representativeness of the datasets that're available.
- Sensor measurements may be affected by calibration issues, environmental interference, sensor drift and hardware limitations.
- Network connectivity can affect the real-time transmission of IoT data.
- The proposed system is a research and development solution. Therefore needs further validation before full commercial-scale deployment.
- Different processing environments may require recalibration of sensors, thresholds and AI models.
- Automated recommendations and alerts depend on the availability and reliability of the underlying sensor data and model outputs.
- The project focuses on the defined fish processing workflow and does not aim to automate every activity inside a commercial fish-processing facility.

3 OVERALL SOLUTION STRUCTURE

3.1 Integrated solution overview

The proposed solution is designed as one integrated AI- and IoT-based Maldive fish processing framework of four separate systems. The framework links the stages of the processing lifecycle starting with raw fish quality assessment and continuing through boiling and salting drying and the final dried-fish quality evaluation.

The central value proposition of the system is to give data-driven quality and process support across the entire production workflow. Each component solves a processing problem while generating information that is useful for other stages of the workflow. This lets the system move from inspection activities toward a connected processing-management approach.

The integrated solution consists of four functional components:

1. AI-Driven Raw Fish Quality Assessment and Standardization – evaluates the quality of fish before processing.
2. Smart Optimization of Boiling and Salt Distribution – supports controlled boiling and consistent salt distribution.
3. Intelligent IoT-Based Multi-Parameter Environmental Monitoring – monitors conditions during drying and provides alerts when conditions become unsuitable.
4. AI-Based Multi-Sensor Quality Evaluation of Dried Maldive Fish – evaluates the product using computer vision and sensor-based measurements.

The four components are linked through a shared data and monitoring structure. The system uses AI models, IoT sensors, microcontrollers, data communication and a centralized dashboard to support monitoring and decision-making. The research architecture includes ESP32-based sensor communication, image acquisition, AI model predictions, sensor-data transmission and centralized visualization.

The overall outcome is intended to be a consistent, traceable and data-driven Maldive fish processing workflow, where information collected at one stage can support decisions at later stages.

3.2 High level architecture

The high-level architecture of the proposed system can be represented as five main layers:

1. Operational Layer

The main users include fish producers, processing-factory operators, quality inspectors, small and medium-scale fish-processing businesses and seafood exporters. These users interact with the system through the monitoring and decision-support interface.

2. Data Acquisition Layer

This layer collects information from cameras and physical sensors. The system uses image acquisition for quality assessment and IoT sensors for process and environmental measurements. ESP32-based communication is used for sensing while other processing parameters are captured through the relevant sensing mechanisms.

3. Intelligent Processing Layer

The collected data are processed by the four components. Computer-vision models perform classification sensor data are processed to identify relevant conditions and process parameters are used for optimization and decision support.

4. Communication and Integration Layer

The components communicate their outputs through the common system architecture. IoT sensor data are transmitted to the monitoring platform while AI components provide prediction and classification outputs. This layer enables information from components to contribute to the overall workflow.

5. Monitoring and Decision-Support Layer

A centralized dashboard provides users with quality results, sensor readings, processing information, alerts and AI predictions. This provides an interface, for monitoring the processing lifecycle.

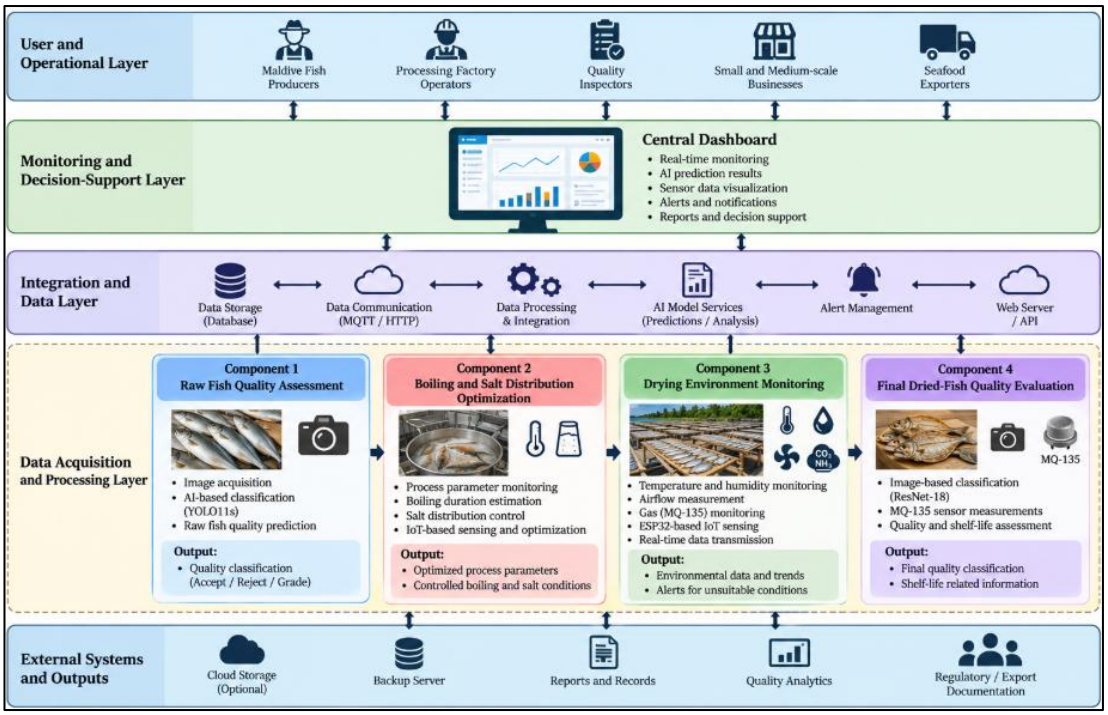


Figure 1 - High-Level Architecture of the Integrated Smart Maldive Fish Processing System

The architecture shows that the components are linked in a sequence but each component does not rely entirely on the others for every operation. Each component carries out a task while giving its output to the shared monitoring and decision-support system.

3.3 Component 1 summary

Component 1 – AI-Driven Raw Fish Quality Assessment and Standardization

Component Owner: Madurawala M D S S – IT22300096

Subproblem:

Component 1 tackles the challenge of consistently checking the quality of raw fish before it goes into the processing line. Manual checks can be biased and slow creating a risk that quality decisions may differ.

Input:

Component 1 receives pictures of fish for visual inspection and classification.

Core Method:

Component 1 uses AI computer vision to spot and classify raw fish quality. The research uses the YOLO11s model for this assessment.

Output:

Component 1 produces a quality classification or prediction that can decide if the fish is ready for further processing.

Contribution to Overall Solution:

Component 1 sets the quality standard of the material at the start of production. Its output gives a quality reference before the fish is boiled, salted, dried and finally inspected. Component 1 therefore acts as the quality check in the integrated system.

3.4 Component 2 summary

Component 2 – Smart Optimization of Boiling and Salt Distribution

Component Owner: Ramesh J N G A – IT22341686

Subproblem:

Component 2 addresses boiling and salt distribution during processing. Long boiling can damage fish structure and hurt efficiency while poor salt control can lead to uneven product quality.

Input:

Component 2 uses data such as fish traits, batch size, boiling conditions, salt level, temperature and related sensor readings.

Core Method:

Component 2 applies sensing and process optimization to help choose and monitor suitable boiling and salting conditions. The research describes a model that estimates boiling time based on fish thickness batch mass and raw-fish quality and also controls brine concentration and core temperature.

Output:

Component 2 gives optimized or controlled process settings and information about boiling and salt distribution.

Contribution to Overall Solution:

Component 2 forms the controlled processing stage between raw-fish assessment and drying. Consistent processing at this stage supplies a better-controlled input for the next drying step.

3.5 Component 3 summary

Component 3 – Intelligent IoT-Based Multi-Parameter Environmental Monitoring

Component Owner: Bandara C S – IT22340214

Subproblem:

Component 3 solves the lack of continuous monitoring of environmental conditions during drying. Changes in temperature, humidity, airflow and gas levels can affect the drying environment.

Input:

Component 3 receives readings from multiple sensors.

Core Method:

An ESP32-based IoT system collects and sends data. The research uses a DHT22 sensor for temperature and humidity a hot-wire anemometer for airflow speed and an MQ-135 sensor for ammonia and amine gases. Sensor data is sent through MQTT for monitoring and analysis.

Output:

Component 3 supplies environmental readings and triggers alerts when drying conditions become unsuitable.

Contribution to Overall Solution:

Component 3 provides the environmental monitoring stage of the workflow. It lets operators see conditions that influence drying and react to any environment.

3.6 Component 4 summary

Component 4 – AI-Based Multi-Sensor Quality Evaluation for Dried Maldive Fish

Component Owner: Sanchala W K S – IT22224934

Subproblem:

Component 4 tackles the limits of using manual visual inspection for final dried-fish quality. Visual traits give quality data while gas sensor readings add insight into product condition and shelf life.

Input:

Component 4 receives pictures of dried fish along with sensor data from an MQ-135 gas sensor.

Core Method:

Component 4 uses a deep-learning model to classify dried-fish quality from images and combines this with MQ-135 sensor data to assess quality and shelf life. The research used 1,200 labeled images split into training, validation and test sets.

Output:

Component 4 delivers a dried-fish quality classification together, with sensor information that supports assessment of product condition and shelf life.

This component is the quality evaluation stage it gives information about the condition of the processed product after boiling, salting and drying.

3.7 End to end user and data flow

The integrated workflow begins when raw fish enters the processing facility. The first stage captures information about the fish and passes it to Component 1 for AI-assisted quality assessment. The resulting classification gives an indication of raw-material quality and helps decide whether to proceed with processing.

The accepted raw fish then moves to the boiling and salting stage. Component 2 receives processing information and sensor measurements and uses the available process model to support appropriate boiling and salt-distribution conditions. The resulting processed fish is then transferred to the drying stage.

During drying Component 3 continuously collects information using the IoT sensor network. Temperature, relative humidity, airflow and gas-related measurements are transmitted for monitoring. When environmental conditions move outside the defined operating conditions the system can generate an alert so that the operator can take an action.

Once the drying process is completed the product enters the quality evaluation stage. Component 4 captures images of the dried fish. Obtains sensor measurements. The ResNet-18 model evaluates quality while MQ-135 measurements provide complementary sensor information. These outputs are used to support the quality and shelf-life-related assessment.

The results generated by the four components are made available through monitoring and decision-support environment. Users can therefore obtain information about raw-material quality, processing conditions, drying conditions, alerts and final product quality within the overall system.

4 COMPONENT INTEGRATION AND COHESIVE OPERATION

4.1 Integration strategy

The proposed system follows an integrated architecture in which the four research components keep their individual functions while communicating through common data and monitoring mechanisms. The integration strategy is based on connecting the components according to the sequence of Maldive fish processing: raw-fish assessment, boiling and salting drying and final dried-fish evaluation.

The project presentation identifies five main integration points: ESP32 sensor communication, image uploads and camera-based image capture, sensor-data transmission AI model predictions and the central dashboard.

The integration approach can be summarized as follows:

1. Component-level development – Each student. Evaluates the assigned component independently.

2. Common integration interfaces – Component outputs are structured so that relevant quality, process, sensor and prediction information can be transferred to the common monitoring environment.

3. IoT communication – ESP32-based sensing is used for sensor-data acquisition and communication in the components.

4. AI output integration – Predictions generated by the AI models are treated as decision-support information that can be displayed alongside sensor and process information.

5. Central monitoring – A central dashboard provides an interface for viewing information generated by the four components.

6. End-to-end validation – Integration is evaluated by considering whether information can move through the processing workflow and whether the combined outputs support the intended decision-making process.

The research paper describes the architecture as an interconnected four-tier pipeline that follows the industrial processing workflow from raw intake through optimized boiling and salting drying-environment monitoring and final product assessment.

Integration Sequence

The primary sequence is:

Raw fish assessment → Boiling and salt-distribution optimization → Drying-environment monitoring → dried-fish quality evaluation → Centralized monitoring and reporting

This sequence reflects the physical movement of the product while the digital system maintains information about each processing stage.

Integration risks are addressed primarily through separation, defined component responsibilities, sensor-data transmission AI prediction outputs and centralized visualization. This reduces the risk that failure or modification, in one component will require redesign of the other components.

4.2 Interfaces data contracts and shared models

The available project documentation identifies the integration interfaces but does not specify a finalized REST API specification, formal JSON schema, versioned endpoint list or complete database schema. Therefore the common report should not claim that such detailed contracts were implemented unless they are documented in the implementation evidence.

Based on the project architecture the principal shared interfaces are shown below.

Table 4 - Integrated System Interfaces

Interface	Source	Destination	Main Data/Output	Purpose
Image input interface	Camera / user upload	Component 1 or Component 4	Fish image	AI-based quality assessment
ESP32 sensor interface	Sensors + ESP32	Monitoring/integration layer	Sensor readings	Collection of environmental/process data
Sensor-data transmission	IoT components	Central monitoring system	Time-stamped sensor data	Remote monitoring and analysis

AI prediction interface	AI models	Central dashboard	Quality/prediction results	Display and decision support
Dashboard interface	Integration layer	Users	Results, readings, alerts, reports	Centralized monitoring
Alert interface	Monitoring/processing logic	Users/operators	Warning notification	Response to unsuitable conditions

Shared Data Concepts

Shared Data Concepts require that every major output contains context to tell one observation apart from another. At a minimum the common data structure must keep:

- processing/batch identification;
- component or processing-stage identification;
- measurement or prediction value;
- timestamp;
- sensor/model information;
- quality or alert status where applicable.

For sensor data the research architecture uses telemetry. For example the drying component sends measurements through an MQTT-based IoT architecture and it creates validated data packets from the sensor readings.

Data Validation

Data that comes from sensors or AI models must be checked before it is shown as system information. Proper validation includes:

- checking whether sensor values are within the supported measurement range;
- identifying incomplete readings;
- associating measurements with the correct processing stage;
- maintaining timestamps for observations;
- distinguishing AI predictions from raw sensor measurements;

Error-Handling Responsibility

The component that creates the data should own any errors. For example:

- Sensor-level errors: handled by the IoT component.;
- Image/input errors: handled by the relevant image-acquisition component.;
- AI prediction errors: handled by the corresponding AI component.;
- Communication failures: identified by the communication/integration layer.;
- Dashboard/display failures: handled by the central monitoring application.

Keeping responsibilities separate gives ownership and lets the central dashboard show the overall system status.

4.3 Shared nonfunctional requirements

The integrated system needs non-functional features so that all four parts can work together as a single solution.

Performance

The system must send sensor data, AI predictions, dashboard updates and alerts enough to help processing decisions. This is very important for monitoring because a delay can make an alert less useful. The drying-monitoring example in the research uses MQTT and continuous telemetry which shows the need for fast data transfer.

Usability

The central dashboard must display quality results, sensor information, predictions and alerts in a way that operators and inspectors can understand. Users should not have to touch the underlying AI models or sensor hardware during use.

Reliability

The system should keep collecting and processing data even if some sensor readings are missing for a time. When possible the system must flag bad readings instead of treating them as good. The research example uses repeated sensor measurements and validated telemetry to make environmental monitoring more reliable.

Security

All communication between sensing devices, processing services, storage and the dashboard should be protected from access. User access should match the responsibilities of operators and inspectors. The Maldive fish project sources do not give proof of completed authentication, authorization or encryption. Therefore treat these as system-level needs, not finished implementations.

Maintainability

The modular design lets an AI model, sensor or processing part be updated without redesigning the whole system. This is important because AI models and sensor calibration may need changes.

Scalability

The design should allow sensors processing stages, quality indicators or monitoring functions to be added later. The current research shows a four-component design; larger commercial deployments would need validation.

Professional Use

The dashboard should use labels easy-to-understand alerts and good visual displays of measurements and predictions. Quality decisions must stay clear, to processing staff of showing only raw technical numbers.

4.4 Integrated workflow and exception handling

Normal End-To-End Operation

Under operation the integrated workflow proceeds through the following stages:

Step 1 – Raw Fish Intake

Raw fish images are taken and sent to Component 1. The AI model looks at the images. Gives a prediction of raw fish quality.

Step 2 – Processing Decision

The raw fish quality result is used for the stage. Component 2 takes batch and process data. Decides and watches suitable boiling and salting conditions.

Step 3 – Boiling and Salting

The component watches boiling conditions, salt level and temperature. It gives optimized or controlled data before the fish goes to drying.

Step 4 – Drying

The fish goes to the drying stage. Component 3 keeps an eye on conditions with its IoT sensors. The setup records temperature, relative humidity, airflow and MQ-135 gas readings.

Step 5 – Environmental Alert

When the drying conditions go beyond the set limits an alert is sent so the operator can act. The rule says that high humidity with low airflow for a long time will trigger an alert.

Step 6 – Final Product Evaluation

After drying Component 4 checks the fish with image classification and MQ-135 sensors. The quality and shelf-life data is then shown in the integrated monitoring system.

Step 7 – Dashboard and Reporting

The main dashboard shows the predictions, sensor data, alerts and quality info to the user

Table 5 - Exception Handling

Exception	Possible Cause	Expected Handling
Invalid image	Poor image quality or unsuitable input	Request a new image or flag the input as invalid
Missing sensor reading	Sensor/network interruption	Mark the reading unavailable and continue monitoring when data resumes
Abnormal environmental condition	High humidity, low airflow, or other unsuitable condition	Generate an alert for operator action
Communication failure	Wi-Fi/network interruption	Record the communication failure and resume transmission when connectivity returns
AI prediction unavailable	Model/service failure	Flag the prediction as unavailable rather than presenting an invalid result
Out-of-range sensor value	Sensor fault or abnormal condition	Validate the value and flag it for inspection
Dashboard unavailable	Application/service failure	Preserve locally available data where supported and restore visualization after recovery

4.5 Team contribution and dependency matrix

Table 6 - Team Contribution and Dependency Matrix

Student	Student ID	Assigned Component	Main Research/Development Responsibility	Main Integration Dependency
Madurawala M D S S	IT22300096	Component 1 – AI-Driven Raw Fish Quality Assessment and Standardization	Raw fish quality assessment and AI-based classification	Provides raw-fish quality information for the processing workflow
Ramesh J N G A	IT22341686	Component 2 – Smart Optimization	Boiling/process optimization and salt-distribution control	Uses relevant raw/process information

		of Boiling and Salt Distribution		and provides controlled processing outputs
Bandara C S	IT22340214	Component 3 – Intelligent IoT-Based Multi-Parameter Environmental Monitoring	IoT sensing and drying-environment monitoring	Provides drying-condition data and environmental alerts
Sanchala W K S	IT22224934	Component 4 – AI-Based Multi-Sensor Quality Evaluation of Dried Maldive Fish	Final product AI classification and sensor-based quality evaluation	Consumes the processed/dried product and provides final quality information

5 INTEGRATED IMPLEMENTATION AND PROJECT EXECUTION

5.1 Common technology stack and infrastructure

The proposed smart Maldive fish processing system uses a mix of Artificial Intelligence (AI) computer vision, Internet of Things (IoT) sensor technologies, microcontrollers, data communication and web-based monitoring.

The technology stack is arranged so that each component can carry out its special tasks while sending its results to the shared monitoring and decision-support environment.

The project architecture highlights ESP32 sensor communication, image capture and upload, sensor-data transmission AI model predictions and a central dashboard as the shared integration technologies.

Table 7 - Common and Component Level Technology Stack

Technology / Platform	Application in the Integrated System
Python	AI/ML development, image processing, data processing, and model implementation
YOLO11s	Raw fish visual quality assessment in Component 1
ResNet-18	Dried-fish image classification in Component 4
ESP32	IoT sensor interfacing and environmental/process data transmission
MQ-135	Gas/VOC-related monitoring in relevant quality and environmental monitoring stages
DHT22	Temperature and relative-humidity measurement
Camera	Image acquisition for AI-based fish quality assessment
MQTT	IoT telemetry/data communication in the environmental-monitoring architecture
REST API / Web services	Communication between AI processing, sensor services, database, and dashboard where implemented
Central Dashboard	Common visualization of predictions, sensor readings, alerts, and quality information
Database / Data Storage	Storage of sensor measurements, predictions, and historical processing information

The detailed dried-fish component architecture describes a Flask-based REST API that acts as a communication hub between the AI model, IoT devices, database and web application. It also shows deployment and relational storage for historical sensor data.

For raw-fish assessment the implementation evidence explains Python and the Ultralytics YOLO11s framework with deployment experiments using a Raspberry Pi 4 and an RGB camera for edge-based inference.

The technology choices are appropriate to the project because they provide:

- support for image classification;
- low-cost microcontroller-based sensor acquisition;
- wireless IoT communication;
- real-time or near-real-time monitoring;
- modular integration of different AI models and sensors;
- a common interface for users to view system outputs.

The project proposal also identifies ESP32 hardware, camera modules, gas sensors and other sensing equipment as part of the planned hardware infrastructure.

5.2 Development and integration process

The project followed a component-based development approach, where each student handled a research and development component while the group kept a common integrated architecture. This approach let the four components be developed independently while defining integration points for communication and data exchange.

The development process can be summarized into the following stages:

1. Problem Identification and Requirement Definition

The group first identified the limitations of conventional Maldive fish processing and split the overall problem into four related subproblems. Each student then defined the research and development requirements for their assigned component.

2. Component-Level Research and Development

Each component was developed according to its technical requirements. These included AI model development for image-based assessment, process optimization for boiling and salting IoT sensor monitoring for drying and multi-sensor final-product assessment.

3. Interface and Integration Definition

ESP32 sensor communication; image upload and camera capture; sensor-data transmission; AI model predictions; and the central dashboard.

4. Component Testing

Each component was evaluated using its relevant performance measures before being considered part of the integrated system. For example AI components use classification-related metrics while the environmental-monitoring component considers sensor and communication performance.

5. Integration

The component outputs were connected to the common monitoring architecture. The integration focused on ensuring that information generated at stages could be transferred to the central monitoring environment.

6. System-Level Evaluation

The final stage considered the workflow rather than only the performance of individual models. This included the movement of information from assessment, through processing and drying to final product evaluation.

Development Management

The available project sources show support for component-based development and clear integration points. Still there is no proof of formal sprint planning Git branching strategy, continuous integration pipelines, code reviews or issue-tracking workflows in the project files that are currently available. Therefore, we should not claim these practices as finished unless the team can provide evidence from MS Planner, Teams, Git or other project-management tools.

5.3 Deployment and operational configuration

The proposed system is built around a mix of edge devices IoT sensing, AI processing, communication services, data storage and a central dashboard.

At the level sensors are linked to microcontrollers like ESP32 devices. These devices gather processing measurements and send the data using the proper communication method. The research work on drying uses MQTT telemetry to send measurements. Draft research paper

For AI image assessment images can be taken with cameras. Uploaded through the application interface. The raw-fish implementation shows edge experiments that use a Raspberry Pi 4 with an RGB camera enabling local image processing and prediction.

The dried-fish quality component explains a backend service that links the AI model, IoT devices, database and web application. The backend receives image data runs model inference gathers sensor measurements and sends results to the dashboard.

Operational Flow

The operational configuration can therefore be summarized as:

Sensors / Cameras → Edge or Processing Device → Communication Layer → AI/Data Processing → Database → Central Dashboard → User

The dashboard gives access to quality predictions, sensor readings and monitoring information. Historical measurements can also help with trend analysis and reporting when the relevant component has stored the data.

Monitoring

Operational monitoring should focus on:

- sensor connectivity;
- incoming sensor-data availability;
- AI model availability;
- successful image processing;
- data transmission;
- dashboard availability; and
- alert generation.

Recovery

The project sources show database and data-storage functions. They do not give enough proof of a finished backup schedule, disaster-recovery plan or tested restoration process for the integrated system. Therefore these should be seen as requirements for a production launch not as finished features.

In a production launch we should add regular database backups, configuration backups, model-version backups and restoration testing.

5.4 Evidence of team execution

[Guidance - delete before submission: Summarize evidence from MS Planner, Teams, Git history, meeting records, and research diaries. Do not paste raw activity logs into the chapter.]

6 INTEGRATED TESTING EVALUATION AND VALIDATION

6.1 Evaluation framework and success criteria

The integrated evaluation framework was created to decide if the proposed system can carry out its intended tasks throughout the fish processing lifecycle while keeping AI performance, process efficiency, sensor reliability and final-product assessment at acceptable levels.

The evaluation looks at both component-level performance and overall system results. Detailed model development and component experiments are recorded in the individual theses while this shared report highlights evidence that shows how the four components contribute to the integrated solution.

The research paper shows evaluation over 20 full-scale production batches each with 15 kg of raw tuna carried out over four months at a commercial processing facility in southern Sri Lanka.

Group-Level Evaluation Questions

1. Can the system reliably assess raw fish quality before processing?
2. Can intelligent process control improve boiling efficiency. Make salt distribution more consistent?
3. Can the IoT subsystem reliably monitor drying environment conditions. Produce timely alerts?
4. Can the final product be evaluated by using AI classification and sensor info?
5. Can the four components give outputs throughout the entire processing workflow?

Table 8 - Evaluation Framework

Evaluation Area	Main Metric / Evidence	Success Criterion
Raw fish assessment	mAP@0.5, precision, recall, F1, inference latency	Reliable classification with real-time-capable inference
Boiling optimization	Boiling time, energy consumption	Improvement compared with conventional/manual processing
Salt distribution	Salt penetration coefficient of variation	Lower variation than conventional processing
Drying monitoring	Telemetry delivery rate, alert response time	Reliable data transmission and timely alerts
Drying performance	Dehydration duration	Improved/stabilized drying performance
Final quality assessment	Accuracy, precision, recall, F1	Reliable classification of dried-fish quality
Integrated operation	Component outputs, data flow, dashboard	Successful transfer and presentation of relevant outputs

6.2 Integration and end to end testing

The integrated testing approach looks at how the four components work with the shared monitoring architecture. The main integration points are image acquisition, sensor communication, AI predictions, sensor-data transmission and the central dashboard.

Interface Testing

Interface testing checks whether:

- image inputs can be supplied to the AI component;
- sensor measurements can be acquired from the relevant IoT devices;
- sensor information can be transmitted to the monitoring environment;
- AI predictions can be represented as system outputs; and
- relevant information can be displayed through the central dashboard.

The project architecture clearly names ESP32 sensor communication, image uploads/camera capture, sensor-data transmission AI model predictions and the central dashboard as integration points.

System Testing

System-level testing follows the physical processing sequence:

fish → Quality assessment → Boiling and salting → Drying monitoring → Final product assessment.

The goal is to confirm that each stage can do its job and that the output from one stage is fit for the stage.

Performance Testing

The research gives evidence for several parts.

For raw-fish inspection YOLO11s reached a 95.8 % mAP@0.5, with an edge inference latency of 16.2 ms (61.7 FPS) in the experiment.

For boiling and salting the optimized process was compared with manual processing across 20 batches. The mean boiling time fell from 58.4 ± 6.2 minutes to 42.1 ± 1.8 minutes while energy use dropped from 3.82 ± 0.24 MJ/kg to 2.94 ± 0.11 MJ/kg.

For drying monitoring the ESP32 node recorded 10,080 one-minute filtered telemetry packets over seven days. Achieved 99.6 % MQTT packet delivery. During a reported rain event an alert was triggered within 2.1 seconds.

Regression Testing

Regression testing matters when individual models, sensor thresholds or dashboard functions are changed. The current project evidence does not provide a formal regression-test log or complete regression-test results. Therefore this should be recorded as an integration-testing activity where the team has matching test records.

Security Testing

The integrated project documentation does not give proof of a finished system-wide security test covering authentication, authorization, encryption and penetration testing. Therefore security should be seen as an identified requirement and limitation rather than a completed security validation.

Usability and User Acceptance Testing

The available research paper shows experimental validation but does not give enough proof of a formal user-acceptance study involving a set number of operators, inspectors or processing-facility users. Therefore formal user-acceptance results should not be claimed without the team's evaluation records.

6.3 Overall results

The available experimental evidence shows results across the main processing stages.

Table 9 - Synthesized Integrated Results

Component / Stage	Evaluation	Reported Result
Raw fish assessment	YOLO11s overall mAP@0.5	95.8%
Raw fish assessment	Edge inference latency	16.2 ms
Raw fish assessment	Edge processing speed	61.7 FPS
Boiling optimization	Reduction in boiling cycle time	27.9% ± 1.4%
Boiling optimization	Reduction in thermal fuel consumption	23.0% ± 1.2%

Salt distribution	Salt variation	18.6% ± 2.1% → 5.2% ± 0.8%
Drying monitoring	MQTT telemetry delivery	99.6%
Drying monitoring	Alert response time	2.1 s
Drying performance	Dehydration duration	4.2 ± 0.3 days vs. 6.5 ± 1.4 days
Final dried-fish assessment	Overall classification accuracy	94.0%

6.4 Stakeholder and user validation

Stakeholder validation is important because the proposed system is meant to help real processing operators, quality inspectors and production managers not just to serve as an experimental AI platform.

The available project research shows validation using production batches and a commercial processing environment. The research reports that the integrated platform was tested across 20 full-scale production batches at a processing facility, in southern Sri Lanka.

However, the current evidence does not provide a formal user-validation study that specifies:

- the number of participants;
- participant roles;
- questionnaire/interview methodology;
- usability scores
- user-acceptance criteria;
- gathered feedback;
- changes carried out specifically after stakeholder feedback.

6.5 Achievement of common objectives

Table 10 - Achievement of Common Objectives

Objective	Evidence	Achievement
1. Develop an AI-assisted raw fish quality assessment mechanism	YOLO11s achieved 95.8% overall mAP@0.5 on the reported test set	Fully achieved based on reported technical evaluation
2. Develop an intelligent boiling and salt-distribution optimization mechanism	27.9% ± 1.4% reduction in boiling cycle time and salt variation reduced from 18.6% ± 2.1% to 5.2% ± 0.8%	Fully achieved based on reported experimental evidence
3. Develop IoT-based multi-parameter drying monitoring and alerting	99.6% MQTT delivery, 2.1 s alert response, environmental monitoring over seven days	Fully achieved based on reported technical evaluation
4. Develop final dried-fish quality evaluation using AI and sensor information	ResNet-18 final-product evaluation reported 94.0% accuracy	Fully achieved based on reported technical evaluation
5. Integrate the four components into a common monitoring and decision-support framework	Integrated architecture, identified interfaces, component outputs, and central dashboard	Partially to substantially achieved; full end-to-end integration evidence should be strengthened

6.6 Integrated limitations and threat

1. Dataset Dependence

AI performance relies on the quality and representativeness of training and testing datasets. The raw-fish model was assessed with 381 test frames while the dried-fish evaluation used a test split of 180 finished dried loins.

These datasets may not fully represent all fish varieties, lighting conditions, processing environments, camera configurations or product conditions encountered across commercial facilities.

2. Limited Production-Site Validation

The reported experiments were conducted across 20 production batches at a processing facility in southern Sri Lanka.

Although this provides operational evidence further testing across multiple facilities, seasons, equipment configurations and production scales would be needed to establish broader generalisability.

3. Sensor Limitations

Gas and environmental sensors can be affected by calibration, environmental interference, sensor drift and changes in operating conditions. Consequently sensor thresholds should be. Recalibrated periodically before long-term commercial deployment.

4. Inter-Component Dependency

The components are logically linked through the processing workflow. Errors in one stage can affect stages. For example incorrect raw-fish classification can conflict with the processing input. Create inconsistencies downstream.

5. Communication Dependency

The IoT components depend on communication for transmitting sensor data. Although the reported drying experiment achieved 99.6 % MQTT delivery the remaining transmission failures show that network communication cannot be assumed to be perfect.

6. AI Generalizations

A high test-set performance does not guarantee the level of performance in every commercial setting.

7. Limited Formal User Validation

The available evidence gives production validation but does not fully document a formal user-acceptance study.

8. Security and Data Protection

The available evidence does not provide a system-wide security evaluation.

9. Commercial-Scale Deployment

The reported research demonstrates technical viability rather than proven large-scale commercial deployment. Additional testing is required to assess long-term maintenance, sensor replacement, hardware durability, operating costs, model updates and integration, with existing factory processes.

7 COMMERCIALISATION SUSTAINABILITY AND RESPONSIBLE INNOVATION

7.1 Value proposition and beneficiaries

The proposed AI and IoT-based Smart Maldive Fish Processing System adds value by making Maldive fish production more consistent, efficient, traceable and easier to monitor. Traditional methods rely heavily on quality checks and operator judgement, which can produce uneven quality choices that require more inspection work and make it hard to keep track of processing and drying conditions all the time. The integrated system solves these problems with automated quality assessment, IoT monitoring, process optimization and a central decision-support dashboard.

The primary beneficiaries are fish producers, processing factories, quality inspectors, small and medium-scale seafood businesses and seafood exporters. The system can help processing plant managers and quality-control supervisors by giving consistent grading, cutting down manual inspection work, helping with quality-related decisions and giving real-time operational data. Quality inspectors and operators can use the dashboard, and automated inspection results in spotting quality problems

The proposed final-quality evaluation component adds automated Grade A, Grade B and Reject decision support by mixing inspection with extra sensor data. The system keeps records like sample identification, timestamps, visual classifications, sensor readings and final quality grades which builds a foundation for quality traceability.

At the industry level the integrated solution can provide benefits through:

- reducing reliance on manual inspection
- improving consistency in quality evaluation
- supporting real-time monitoring of environmental and processing conditions
- spotting poor-quality products before distribution
- supporting data-driven production decisions
- keeping digital records of inspection and processing information
- giving a base for future automated sorting and production analytics

The system therefore gives value not only through single AI predictions but also through putting quality information together across the whole processing life cycle.

7.2 Market alternatives and differentiation

The proposed system is mainly for Maldive fish processing facilities in Sri Lanka, small and medium-size enterprises that produce dried seafood export-oriented seafood processing companies, fishery cooperatives, government processing centers and food-quality certification organizations. The main users will be processing plant managers, quality-control supervisors, quality inspectors and production operators.

The main alternatives to the proposed system are manual inspection, camera-only inspection systems and separate sensor-based monitoring systems. Manual inspection depends a lot on the operator's experience and judgment while camera-only systems look mainly at traits. Sensor-only approaches can give data about environmental or gas conditions but cannot judge the visual look of the product by themselves. The component report shows that mixing these information sources is better than using one inspection method

The proposed system stands out by mixing Artificial Intelligence, computer vision, IoT sensing, gas/VOC sensing, environmental monitoring, process optimization and a central dashboard. The proposal points out automated A/B/Reject grading, real-time monitoring, analytics and the mix of sensing technologies as key strengths.

The main differentiation can therefore be summarized as follows:

Alternative	Main Limitation	Proposed System Advantage
Manual inspection	Subjective and slow	AI-assisted and records information in digital form
Visual-only inspection	Limited to visible traits	Combines visual data with additional sensor data
Sensor-only monitoring	Cannot judge visual appearance	Blends sensor data with computer vision
Isolated IoT monitoring	Mostly focuses on environmental/process data	Links monitoring with AI-based quality decisions
Simple pass/fail inspection	Limited quality categories	Supports Grade A, Grade B, and Reject classification
Separate processing tools	Information is spread across different stages	Provides an integrated processing and quality-management framework

7.3 Business model and feasibility

The proposed commercialization uses a hybrid business model that mixes hardware deployment with ongoing software and support services. The proposal lists four ways to earn money: a one-time system installation fee, a software-as-a-service subscription, maintenance and support services and optional custom analytics services.

The proposed model can operate as follows:

1. System installation fee – A single payment for installing hardware setting up sensors, edge-computing devices and launching the AI system.

2. Software subscription – yearly payments for dashboard use AI updates, cloud storage and technical help.

3.. Support – Optional contracts that cover sensor calibration, hardware upkeep, software updates and system upgrades.

4. Custom analytics services – services that analyze production track quality trends examine batches and report operational performance.

This model fits both large processing sites because the system can be added step, by step. A small processor might start with quality monitoring and dashboard tools while larger sites could add processing-monitoring and optimization parts later.

The technical feasibility is supported by the project implementation, which combines available components such as Raspberry Pi, ESP32, camera modules, gas sensors, environmental sensors and software-based AI models. The individual component also demonstrates that image classification, sensor acquisition, data integration and digital quality recording can be implemented as functions.

The project proposal provides a hardware cost structure including a device ESP32, camera, moisture sensor, VOC sensor, lighting, load cell, enclosure, wiring and mounting equipment. The group presentation estimates the project budget at approximately LKR 75,000 indicating that the prototype has been developed with relatively accessible hardware requirements.

However the current cost estimate should be regarded as a prototype/development cost than a validated commercial production cost. A commercial feasibility study would

additionally need to consider installation, calibration, networking, cloud infrastructure, software maintenance, replacement sensors, operator training, cybersecurity, technical support and long-term operational costs.

Therefore the current evidence supports feasibility and a plausible commercialization model but further field trials and cost-benefit analysis are required before making claims about commercial profitability or return on investment.

7.4 Adoption intellectual property and compliance

Several factors must be considered before the system can be adopted in operational food-processing environments. The important adoption requirement is ensuring that AI and sensor outputs are treated as decision-support information rather than as an absolute food-safety authority. The individual research specifically states that the proposed AI system should not replace laboratory-based chemical or food-safety testing and that MQ-135 measurements should not independently be interpreted as an indicator of food-safety spoilage level or product expiry.

Accordingly commercial deployment should maintain oversight and clearly communicate the limitations of the AI and sensor outputs. The system should provide documentation covering the datasets quality-label definitions, model architecture, training procedures, evaluation metrics, limitations, sensor interpretation and decision rules. This reduces the risk of users treating an AI model as an infallible quality authority.

Intellectual Property

The project proposal identifies two areas of potential intellectual property:

- the AI-based multi-sensor grading algorithm
- the integrated IoT and AI system architecture.

In a commercial product, ownership and licensing arrangements should be formally established among the project contributors and the institution before commercialization. Source-code ownership, trained model ownership, dataset rights, hardware designs, system documentation and any third-party software or model licenses should be clearly documented. At the research stage the project documents

identify potential IP areas but do not establish a finalized ownership or licensing agreement.

Data Protection and Responsible Data Management

The system stores information such as sample IDs, timestamps, images, sensor measurements and quality classifications. Such data should be protected against access, modification or loss. Role-based access, authentication, secure communication, controlled data storage and backup procedures should be considered for a production deployment.

The project should also maintain traceability between collected data. Reported results. The individual research emphasizes dataset separation, documented experimental procedures, reporting of actual results, acknowledgement of limitations and traceability of collected and reported information.

Food Safety and Professional Responsibility

The system should be positioned as an decisions support system rather than a replacement for established laboratory testing or applicable food-safety procedures. Before adoption its output should be validated against appropriate reference-quality assessments under real processing conditions. The individual component specifically identifies the need for datasets, representative samples and additional validation before the grading mechanism can be considered suitable for commercial quality-control applications.

Other important adoption barriers include sensor calibration, model generalization, changing lighting and environmental conditions differences between fish batches, operator training, system maintenance and integration with existing factory workflows. Commercial-scale testing using batches and real processing environments is therefore required before full deployment.

7.5 Sustainable Development Goal contribution

The proposed system has contributions to several Sustainable Development Goals (SDGs) particularly SDG 2 – Zero Hunger, SDG 9 – Industry, Innovation and Infrastructure and SDG 12 – Responsible Consumption and Production.

SDG 2 – Zero Hunger

The system can contribute indirectly to SDG 2 by supporting the reduction of food losses during seafood processing. Improved quality inspection, drying monitoring and identification of poor-quality products can help reduce losses and improve the utilization of processed fish resources. The project proposal identifies reduction of post-processing food losses as its SDG 2 contribution.

However, this contribution relies on the system making reliable quality decisions. Incorrectly rejecting fish can create unnecessary waste. Therefore, model validation and proper human oversight are essential to guarantee that automation does not bring losses.

SDG 9 – Industry, Innovation and Infrastructure

The project directly supports SDG 9 by introducing AI, IoT, computer vision, sensor technologies and data-driven decision-making into a seafood-processing environment. The integrated system offers a foundation for modernizing processing activities and for bringing Industry 4.0-oriented practices into Maldive fish production.

The modular architecture also allows expansion by adding more sensors, better AI models, automated sorting, dashboards and integration with other processing stages.

SDG 12 – Responsible Consumption and Production

The strongest sustainability contribution relates to SDG 12. Grading and improved process monitoring can help reduce avoidable product waste, support better packaging decisions and make better use of raw materials. The project proposal highlights waste minimization through grading and packaging optimization as a contribution to SDG

The system also supports production by creating digital records of processing and quality information. These records can be used to spot recurring quality problems and support process improvements.

Nevertheless, the sustainability benefits must be evaluated alongside the operational costs of deploying electronic hardware. Sensors, computing devices, cameras, batteries, power supplies and other electronic components create maintenance needs and eventual electronic waste concerns. Sustainable deployment should therefore include equipment maintenance, component replacement planning, disposal of electronic equipment and efficient use of computing resources.

Overall SDG Contribution

Overall, the proposed system offers a pathway toward more efficient, innovative and responsible Maldive fish processing. Its greatest potential contribution lies in reducing quality losses improving process consistency and supporting the digital modernization of seafood processing. However, these benefits must be confirmed through commercial-scale studies that measure actual reductions in waste energy use, rejected batches and resource consumption rather than being assumed solely from prototype results.

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Overall conclusion

The proposed AI and IoT-Based Smart Maldive Fish Processing System tackles the main problem of inconsistent, manually dependent and insufficiently monitored Maldive fish processing by linking quality assessment, processing optimization, environmental monitoring and final product evaluation within a single technological framework. Then treating raw-fish inspection, boiling and salting drying and final quality assessment as separate activities the proposed solution connects their outputs through shared data flows and a central monitoring platform.

The experimental results show that individual technologies can provide support at different stages of processing lifecycles. The raw-fish assessment achieved a reported mAP@0.5 of 95.8%. The optimized boiling and salting process reduced boiling time from 58.4 ± 6.2 minutes to 42.1 ± 1.8 minutes and lowered salt-penetration variation from $18.6\% \pm 2.1\%$ to $5.2\% \pm 0.8\%$ across 20 experimental batches.

The environmental-monitoring component showed IoT data transmission with 10,080 telemetry packets and 99.6% MQTT delivery. The alert response during a change was 2.1 seconds. The same evaluation reported a reduction in dehydration duration from 6.5 ± 1.4 days under conditions to 4.2 ± 0.3 days under monitored conditions. The final dried-fish assessment further offers a mechanism for evaluating finished products using visual and supplementary sensor information.

Therefore, the project shows that AI, computer vision, sensing and data-driven process monitoring can be combined to support a more consistent and measurable Maldive fish processing workflow. The evidence supports the feasibility and research value of the proposed integrated approach. However, the system should not yet be viewed as a production-ready industrial solution because larger datasets, broader field validation, long-term operation, formal user validation and commercial-scale testing are still required.

8.2 Principal integrated contribution

The main contribution of the project is not merely the development of four technological components. Its primary contribution is the creation of a cyber-physical framework for Maldive fish processing, where information, from different stages can inform common quality and production decisions.

The integrated solution connects:

1. Raw fish quality assessment – evaluates fish before processing.
2. Boiling and salt-distribution optimization – uses process information to improve processing efficiency and product uniformity.
3. Environmental monitoring – continuously observes drying conditions. Provides alerts when abnormal conditions occur.
4. Final dried-fish quality evaluation – combines computer vision and supplementary sensor information to support product grading.
5. Centralized monitoring and data management – brings component outputs together to support real-time monitoring, alerts, quality reports and decision-making.

This integration gives a role than a single AI classifier or a single sensor-monitoring device. It builds a base for digital traceability, where raw material condition, processing conditions, environment facts and final product quality can all be linked in the same production history.

The project shows how useful it is to split an industrial problem into clear working parts. Each part has a job but still shares data with the main system. This modular way lets us improve AI models, add more sensors or bring in extra automation all without rebuilding the whole system.

Another contribution is spotting the limits that must be handled before industry can use it. The project says that sensor data and AI classifications should be seen as help for decision making and should not replace lab-based food quality or safety checks.

8.3 Future directions

Future work should aim to make the evidence stronger the system more reliable, bigger and easier for use. The next steps are listed below.

8.3.1 More Diverse Datasets

The AI models need to be checked with larger data sets that come from many different fish batches, sizes, shapes, lighting, processing places and production sites. Varied data will show if the performance works outside the lab tests. The research clearly says that size and variety of data are key for model reliability.

8.3.2 Multi-Site and Commercial-Scale Validation

The system must be checked in real fish-processing places in the Maldives not just a small lab test. Big-scale trials should look at how reliable it's how big it can grow how much it costs to run what maintenance is needed and how it works in different environments and production states. The research also says that big-scale testing with data and many batches is needed before real use.

8.3.3 Improved Sensor Calibration and Sensor Fusion

More tests should set up links between sensor data and fish-quality values measured separately. In particular the MQ-135 should not be seen as the way to tell if fish is spoiled or expired. Future work should look at calibration steps and when useful add more precise sensors for moisture and volatile chemicals.

The present rule-based multi-sensor method can grow into a machine-learning sensor mix. A bigger dataset that has been tested could let the system learn how visual features, VOC responses, environment facts and real quality marks connect, of relying mostly on hand-made decision rules.

8.3.4 Improved AI Models and Continuous Model Updating

New releases can compare ResNet-18 and YOLO11s with lighter models and look at transfer learning, data boost and multimodal learning. Ongoing model improvement with industrial data can help fix changes in fish look, light, processing and production settings.

8.3.5 Automated Sorting and Process Control

The quality results can later link to machines that sort so that Grade A, Grade B and Reject items go apart automatically. Likewise new releases could tighten the link between AI guesses and processing controls letting the system go from watching and helping decisions to full automation. The research says automated sorting is a goal.

8.3.6 Advanced Dashboard and Traceability

The main dashboard can grow to give batch histories, quality trends, sensor histories, alerts, production numbers and comparison reports. Connecting raw fish results, processing settings drying facts and final quality would let research spot links between how things are processed and the end product quality.

8.3.7 Formal User and Stakeholder Validation

Future checks should have organised tests, with processing workers, quality inspectors, plant managers and other key people. These tests should look at how easy it's to use how clear the quality grades are, how helpful alerts are, how much workers trust it how it fits into daily work and how ready people are to use the system. That would add to the tech tests and show real user approval.

8.3.8 Security, Reliability and Maintainability

Before using it in business the integrated system must get security and reliability checks. New versions should have communication, limited user access, data backup fault spotting, sensor calibration steps, software update ways and recovery plans. These things matter a lot when the system runs all the time in working plants.

8.3.9 Economic and Sustainability Evaluation

I think a future commercial study should compare the cost of system installation and maintenance with benefits such as lowered processing time lowered energy consumption, lower product losses, reduced manual inspection effort and improved product consistency. I also believe that environmental impacts associated with hardware, maintenance and eventual disposal should be evaluated.

8.3.10 Stronger Integration of the Four Components

I think the final long-term direction is to strengthen the relationship between the four components so that the system becomes an end-to-end intelligent processing platform. I also believe that processing conditions optimization results, environmental measurements, raw-fish quality and final product quality should be stored and analyzed together. This would enable models that can identify how decisions made at earlier processing stages influence final Maldive fish quality.

Overall I think the next stage of the research should move from prototype-level demonstration toward larger-scale, multi-site, user-centered and commercially realistic validation. I believe this would provide the evidence required to determine whether the proposed integrated system can be reliably deployed as a smart-processing solution for the Maldive fish industry.

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APPENDICES

[Guidance - delete before submission: Include only material needed to verify the integrated solution, such as interface specifications, contribution evidence, integrated test matrices, approvals, or extended results. Label each appendix and cite it from the text.]

Appendix A Integrated interface specification

[Guidance - delete before submission: Replace this appendix title and content as appropriate.]

Appendix B Team contribution declaration

[Guidance - delete before submission: Provide a concise, agreed record of individual and shared contributions. All four members should confirm it.]

TEMPLATE CONVENTIONS

[Guidance - delete before submission: Delete this entire section before final submission. It is placed at the end so that the official title page remains the first page while you work.]

- Use the built-in Title, Heading 1, Heading 2, Heading 3, Normal, Caption, and Guidance styles. Do not imitate headings using manual font changes.
- Replace and delete all grey italic guidance text and all square-bracket placeholders before submission.
- Use Times New Roman 12 point for body text and 1.5 line spacing. Abstracts, tables, captions, long quotations, and reference entries may use single spacing.
- Retain A4 page size and the prescribed margins: 40 mm left and bottom; 25 mm top and right.
- Start each chapter on a new page. Use lowercase Roman page numbers for preliminary pages and Arabic page numbers for the main body and appendices.
- Create captions using Word's Insert Caption command. Number figures and tables by chapter and refer to each one by number in the text.
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- Use clear academic language, proofread the final document, disclose AI assistance, and verify every citation, quotation, result, and claim.
- Keep the common report focused on the complete solution, component relationships, integration, overall evaluation, and shared value. Do not paste four individual theses into it.